

(TR-104) PYTHON WITH AI/ML

Training Day 26 Report

Date: 16 February 2026

Topic Covered: Resume / Job Description Classifier using Classical NLP

The twenty-sixth day of training focused on building an NLP-based Resume and Job Description Classifier using classical NLP techniques and machine learning models. The project involved preprocessing real-world resume text, vectorization, model training, and evaluation.

- Built a Resume / Job Description Classifier for categorizing resumes into Data Science, Web Development, Marketing, and HR domains.
- Performed dataset preparation and organized resume samples according to categories.
- Implemented text cleaning and preprocessing techniques on resume data.
- Applied TF-IDF vectorization for converting textual data into numerical features.
- Performed train-test split for model training and evaluation.
- Trained classification model using classical NLP and machine learning techniques.
- Evaluated model performance using accuracy, precision, recall, and F1-score metrics.
- Generated confusion matrix for analyzing classification results.
- Implemented a prediction function `predict_resume_category(text)` to classify input resume text.
- Structured the NLP pipeline in a clean and modular manner.
- Used logging techniques for structured output and debugging instead of print statements.
- **GitHub Link:**
<https://github.com/JaspinderKaurWalia26/Resume-Job-Description-Classifier>
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/resume-text-classifier-using-classical-nlp-789587db0bb7>